

Performance Metrics

TTK 4260: Multivariate Data Analysis and ML

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Science and Technology

- Referene group members send me an email
- Take away from first survery
 - Too long assignment → Some assignments will be long some very short as we progress. I will keep in mind from the next time. Today no assignment.
 - Theoretical questions → Students ask about exam questions. I am providing materials as we go so that you have plenty of time to review instead of my providing them at the correct time.
 - How to use the notebooks. Can you specify when to switch to the notebooks in slides → The slides should be sufficient, but if you find anything unbelievable in them, you can test it using the interactive notebooks. It is a favor done by me for you. It is not possible to sit down and refer to all the cell numbers in the slides. :)
 - No video streaming → Previous years (2023 and 2025) videos are provided. I am responsible only for the 2026 lecture and course materials.

Opening: Why Metrics Matter

The Cancer Detection Paradox

Scenario

A hospital builds a cancer detection model.

The model achieves **99% accuracy**.

The board is thrilled. Should they deploy it?

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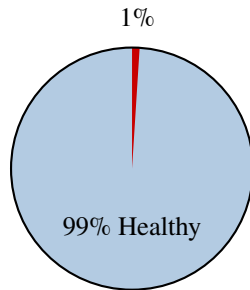
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The Reveal

Dataset: 99% healthy, 1% cancer

Model: Always predicts “healthy”

Catches ZERO cancer cases!



Key Message: The choice of metric can make a terrible model look brilliant.

Part 1: Fundamentals

- Regression Metrics
- Classification Metrics
- Probabilistic Predictions

Part 2: Domain-Specific

- Computer Vision Metrics
- Perceptual & Semantic Losses

Part 3: Advanced

- Distribution Distances
- Ranking & Retrieval Metrics

Throughout:

- **Trap examples** to challenge assumptions
- Practical selection guidelines

Regression Metrics

Core Regression Metrics

Mean Absolute Error (MAE):

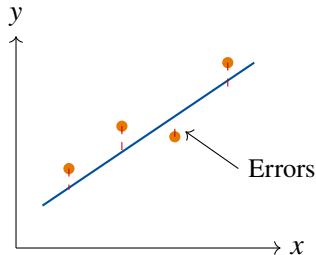
$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$



Trap #1: The Outlier Ambush

Data:

Actual	Predicted	Error
10	11	1
12	11	1
11	12	1
13	12	1
10	11	1
100	11	89

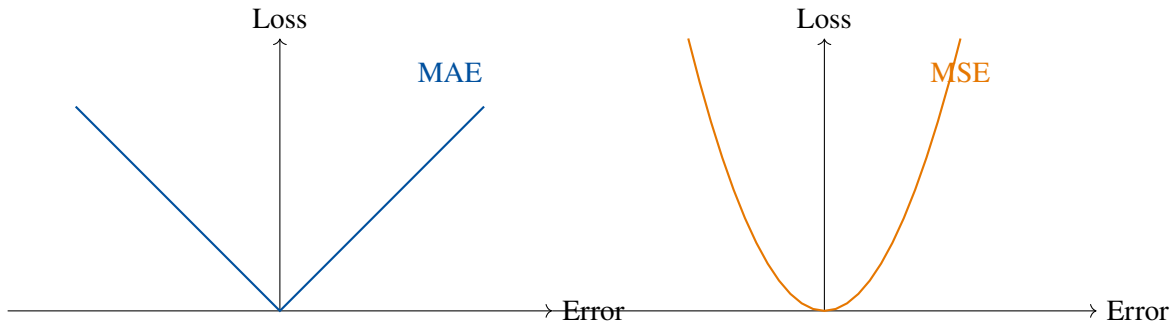
Impact Analysis:

- First 5 errors: $5 \times 1^2 = 5$
- Last error: $89^2 = 7,921$

Metric	Value
MAE	15.67
MSE	1,321
RMSE	36.35

ONE outlier dominates MSE completely!

When to Use MAE vs MSE



Use MAE when:

- All errors matter equally
- Outliers should not dominate
- Example: Delivery time prediction

Use MSE when:

- Large errors are catastrophic
- Need gradient smoothness at 0
- Example: Financial risk, safety

Trap #2: R^2 Can Be Negative!

Common misconception: $R^2 \in [0, 1]$

Reality: $R^2 \in (-\infty, 1]$

Example: Perfectly inverted predictions

Actual	Predicted
1	5
2	4
3	3
4	2
5	1

Calculation:

$$\bar{y} = 3$$

$$SS_{res} = 16 + 4 + 0 + 4 + 16 = 40$$

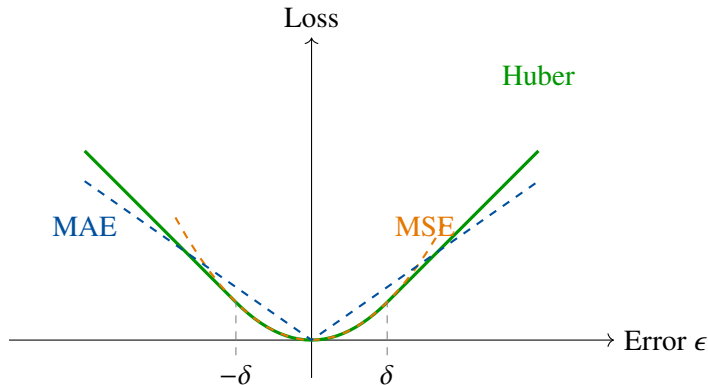
$$SS_{tot} = 4 + 1 + 0 + 1 + 4 = 10$$

$$R^2 = 1 - \frac{40}{10} = \textcolor{red}{-3}$$

Interpretation

$R^2 < 0$ means your model is **worse than predicting the mean!**

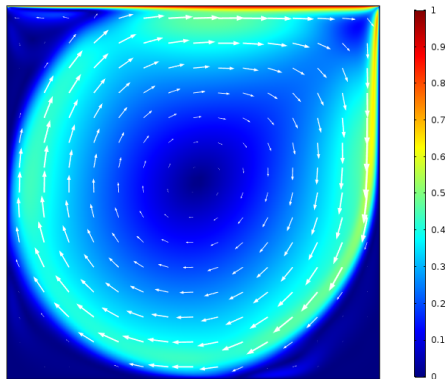
Huber Loss: Best of Both Worlds



$$L_{\delta}(\epsilon) = \begin{cases} \frac{1}{2}\epsilon^2 & \text{if } |\epsilon| \leq \delta \\ \delta(|\epsilon| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

Quadratic near zero (smooth gradient) + **Linear far from zero** (robust to outliers)

Trap: MSE Ignores Low-Magnitude Regions in Multi-Scale Problems



Lid-Driven Cavity Flow

Velocity magnitude varies by **orders of magnitude**

Standard MSE Loss:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{v}_i^{pred} - \mathbf{v}_i^{true}\|^2$$

The Problem — Same 10% relative error:

Region	$ \mathbf{v}^{true} $	Error	MSE contrib.
Lid	1.0	0.1	$(0.1)^2 = \mathbf{0.01}$
Corner	0.001	0.0001	$(0.0001)^2 = 10^{-8}$

MSE is Blind to Low-Velocity Errors!

The lid region contributes $\mathbf{10^6 \times}$ more to the loss than corners - even for the **same relative error**.

Model learns: “Who cares about corners?”

Why This Happens: Scale Sensitivity of MSE

Consider two points with the same 10% relative error:

High-velocity region (lid)

$$v^{true} = 1.0$$

$$v^{pred} = 0.9 \quad (10\% \text{ error})$$

MSE contribution = **0.01**

Low-velocity region (corner)

$$v^{true} = 0.001$$

$$v^{pred} = 0.0009 \quad (10\% \text{ error})$$

MSE contribution = **10^{-8}**

Ratio of contributions: $\text{MSE}_{lid} / \text{MSE}_{corner} = 10^6$

Consequence for Training

- Gradient signal dominated by high-magnitude regions
- Model can have **1000% error** in corners and barely affect the loss
- Physics in low-velocity regions (secondary vortices, boundary layers) is **ignored**

Better Alternatives for Multi-Scale Velocity Prediction

1. Relative/Percentage Error:

$$\mathcal{L}_{rel} = \frac{1}{N} \sum_{i=1}^N \left(\frac{\|\mathbf{v}_i^{pred} - \mathbf{v}_i^{true}\|}{\|\mathbf{v}_i^{true}\| + \epsilon} \right)^2$$

ϵ prevents division by zero (e.g., $\epsilon = 10^{-6}$)

2. Normalized MSE (per-region):

$$\mathcal{L}_{norm} = \frac{1}{N} \sum_{i=1}^N \frac{\|\mathbf{v}_i^{pred} - \mathbf{v}_i^{true}\|^2}{\sigma_{local}^2 + \epsilon}$$

Normalize by local variance

3. Weighted MSE:

$$\mathcal{L}_w = \frac{1}{N} \sum_{i=1}^N w_i \cdot \|\mathbf{v}_i^{pred} - \mathbf{v}_i^{true}\|^2$$

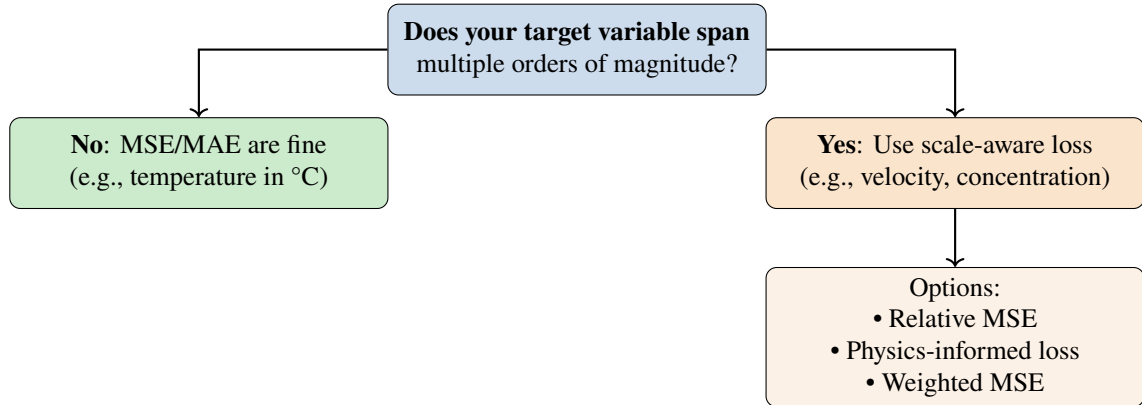
where $w_i \propto \frac{1}{\|\mathbf{v}_i^{true}\|^2 + \epsilon}$

4. Physics-Informed Loss:

$$\mathcal{L} = \mathcal{L}_{MSE} + \lambda_1 \underbrace{\|\nabla \cdot \mathbf{v}\|^2}_{\text{continuity}} + \lambda_2 \underbrace{\|\text{NS residual}\|^2}_{\text{momentum}}$$

Enforce physical constraints regardless of magnitude

Recommendation: Match Loss Function to Physical Meaning



General Principle

Your loss function should match what you care about physically.

Classification Metrics

The Confusion Matrix: Reading the Map

MODEL OUTPUT

ACTUAL (Ground Truth)			
		YES	NO
	YES	TP True Positive Pred: YES Act: YES	FP False Positive Pred: YES Act: NO
	NO	FN False Negative Pred: NO Act: YES	TN True Negative Pred: NO Act: NO

Reading the Matrix

- **Columns** = what is **actually** true
- **Rows** = what the **model predicted**

Correct Predictions

Diagonal = model got it right
(TP and TN)

Errors

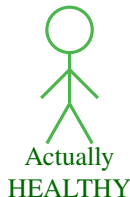
Off-diagonal = model made mistakes
(FP and FN)

False Positive vs False Negative: A Medical Example

Scenario: COVID-19 test where **Positive = Has COVID**

False Positive (FP)

Type I Error



Consequence:

Unnecessary quarantine,
anxiety, wasted resources

False Negative (FN)

Type II Error



Consequence:

Disease spreads,
no treatment given

Key Insight: Which Error is Worse Depends on Context!

COVID test: FN is usually worse (untreated patient spreads disease)

Spam filter: FP might be worse (important email goes to spam)

Key Metrics: What Each One Measures

Precision

*“When I say YES,
am I right?”*

Pred YES →

TP	FP
FN	TN

$$\frac{\text{TP}}{\text{TP} + \text{FP}}$$

Of all **predicted positives**,
how many are actually positive?

Recall (Sensitivity)

*“Did I find all
the actual YESes?”*

Actually YES ↓

TP	FP
FN	TN

$$\frac{\text{TP}}{\text{TP} + \text{FN}}$$

Of all **actual positives**,
how many did I catch?

Specificity

*“Did I correctly
identify the NOs?”*

Actually NO ↓

TP	FP
FN	TN

$$\frac{\text{TN}}{\text{TN} + \text{FP}}$$

Of all **actual negatives**,
how many did I identify?

Trap #3: The Accuracy Paradox

Scenario: Fraud detection where **Positive = Fraud** (1% fraud rate)

Our “Smart” Model:

	Fraud	Normal
Fraud	10 TP	10 FP
Normal	90 FN	9,890 TN

$$\text{Accuracy} = \frac{10+9890}{10000} = 99.0\%$$

$$\text{Recall} = \frac{10}{10+90} = 10\%$$

“Dumb” Model (always says Normal):

	Fraud	Normal
Fraud	0	0
Normal	100 FN	9,900 TN

$$\text{Accuracy} = \frac{0+9900}{10000} = 99.0\%$$

$$\text{Recall} = \frac{0}{0+100} = 0\%$$

Both Have 99% Accuracy!

The “dumb” model catches **ZERO** fraud but looks equally good by accuracy.

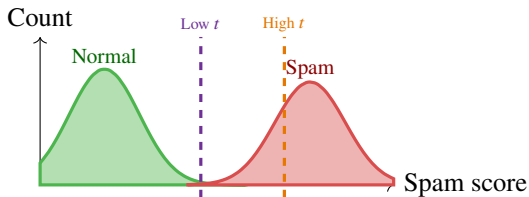
⇒ **In imbalanced data, accuracy is nearly meaningless!**

Trap #4: The Precision–Recall Trade-off

Task: Spam filter where **Positive class = SPAM**

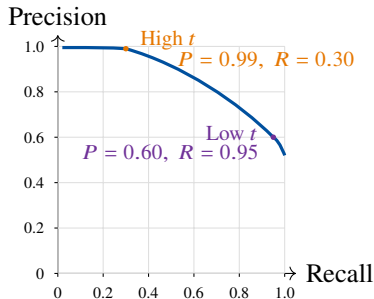
How a Classifier Works

- Model outputs a “spam score” $s(x) \in [0, 1]$
- We choose a threshold t
- Predict SPAM if $s(x) \geq t$



High t : Conservative — few FP, many FN

Low t : Aggressive — few FN, many FP



Key Insight

Same model, different thresholds!

Moving t traces the PR curve.

You cannot have both high P and high R.

Which Threshold is Better? It Depends on Error Costs!

Remember: Positive = SPAM

The Two Error Types for Spam

False Positive (FP):

Normal email → Spam folder

User misses important message!

False Negative (FN):

Spam → Inbox

User sees annoying spam

Context	Best Choice
Business email	High t (conservative) FP very costly — can't lose contracts!
Personal email	Low t (aggressive) FN tolerable — just delete spam
Newsletter	Depends on user Some want everything, some hate spam

F1 Score: Harmonic Mean of Precision and Recall

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

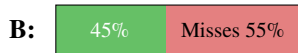
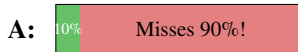
Punishes extreme imbalance: if either P or R is low, F1 will be low.

Trap #5: F1 Score Hides Critical Information

Scenario: Cancer screening where **Positive = Has Cancer**

Model	Precision	Recall	F1
A	90%	10%	18%
B	45%	45%	45%

Which is better for screening?



Analysis:

- Model A: High precision (90%)
BUT catches only 10% of cancers!
- Model B: More false alarms
BUT catches 4.5× more cancers

For Screening Tests

Recall is critical!

Missing cancer (FN) is far worse than a false alarm (FP) that triggers more tests.

Lesson

F1 alone hides **which** errors you're making.
Always examine P and R separately!

ROC Curve: Understanding the Trade-off Visually

ROC = Receiver Operating Characteristic

X-axis: False Positive Rate (FPR)

$$FPR = \frac{FP}{FP + TN} = 1 - \text{Specificity}$$

“What fraction of negatives did I wrongly flag?”

Y-axis: True Positive Rate (TPR)

$$TPR = \frac{TP}{TP + FN} = \text{Recall}$$

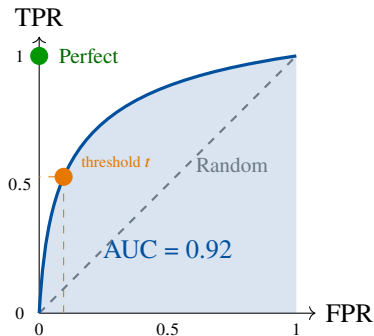
“What fraction of positives did I catch?”

Each point on the curve = one threshold t

AUC = Area Under Curve

● AUC = 1.0: Perfect

● AUC = 0.5: Random guessing



Higher curve = better model
Closer to top-left corner = better trade-off

Trap #6: ROC-AUC Can Mislead with Imbalanced Data

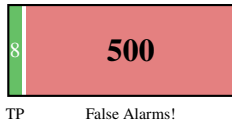
Scenario: Rare disease (0.1% prevalence = 10 sick per 10,000 people)

Model with AUC = 0.95 (sounds great!)

At threshold giving 80% TPR (catches 8 of 10 sick):

From ROC curve: FPR $\approx$ 5%

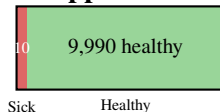
But 5% of 9,990 healthy =



$$\text{Precision} = \frac{8}{8+500} = \mathbf{1.6\%}$$

Only 1 in 63 positive predictions is correct!

Why Does This Happen?



ROC uses **rates**, not **counts**.
5% of a **huge** number is still **huge**!

Solution: Use PR-AUC

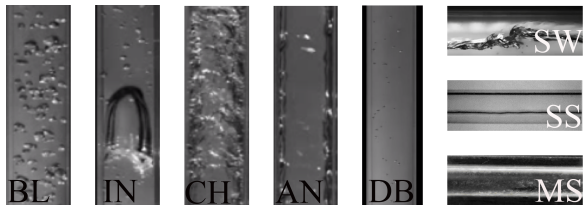
Precision-Recall curve directly shows the false alarm problem in imbalanced data.

Rule of Thumb

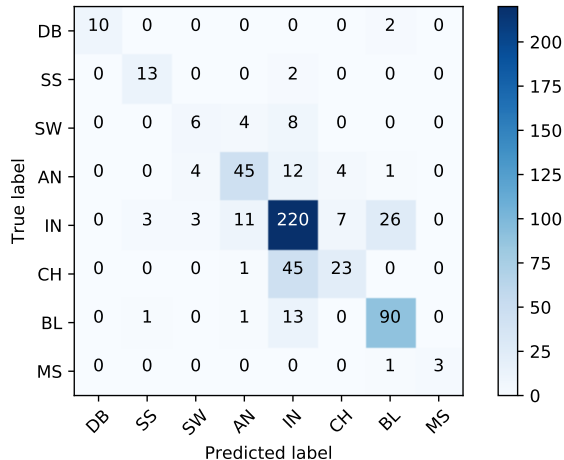
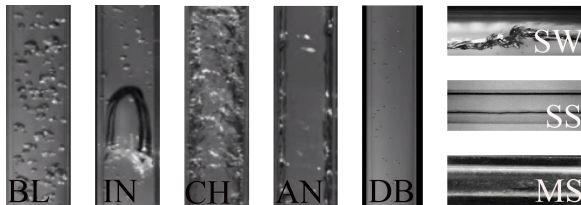
ROC-AUC for balanced data

PR-AUC for imbalanced/rare events

Extending the concepts to multi-class classification



Extending the concepts to multi-class classification



Probabilistic Predictions

Why Hard Predictions Aren't Enough

Scenario: Two models predict “spam” correctly.

Model	Confidence	Correct?	Log Loss
A	51%	✓	0.67
B	99%	✓	0.01

Accuracy treats these identically! But Model B is far more useful.

Log Loss (Cross-Entropy):

$$-\frac{1}{N} \sum [y \log(p) + (1 - y) \log(1 - p)]$$

Severely punishes confident wrong predictions!

Brier Score:

$$\frac{1}{N} \sum (p_i - y_i)^2$$

Like MSE for probabilities.

Range: $[0, 1]$ (lower is better)

Computer Vision Metrics

Signal Quality: SNR and PSNR

Signal-to-Noise Ratio (SNR):

$$SNR = 10 \cdot \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right) \text{ dB}$$

SNR (dB)	Quality
< 10	Very noisy
10–20	Acceptable
20–30	Good
> 40	Excellent

Peak SNR (PSNR):

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX^2}{MSE} \right)$$

For 8-bit images ($MAX = 255$):

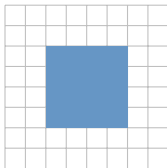
$$PSNR \approx 48.13 - 10 \log_{10}(MSE)$$

Trap #7

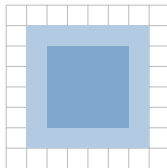
Same PSNR can look very different!

- Blur: PSNR=28dB → “OK”
- Noise: PSNR=28dB → “Terrible”
- 1px shift: PSNR=28dB → “Identical!”

PSNR vs Perceptual Quality

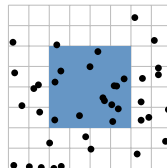


Original



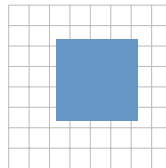
Blurred

PSNR=28dB



Noisy

PSNR=28dB



1px Shift

PSNR=28dB

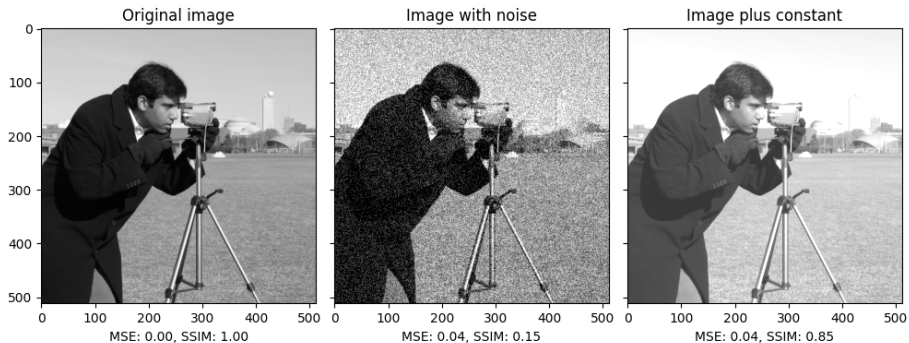
Key Lesson

PSNR measures mathematical similarity, not perceptual similarity!

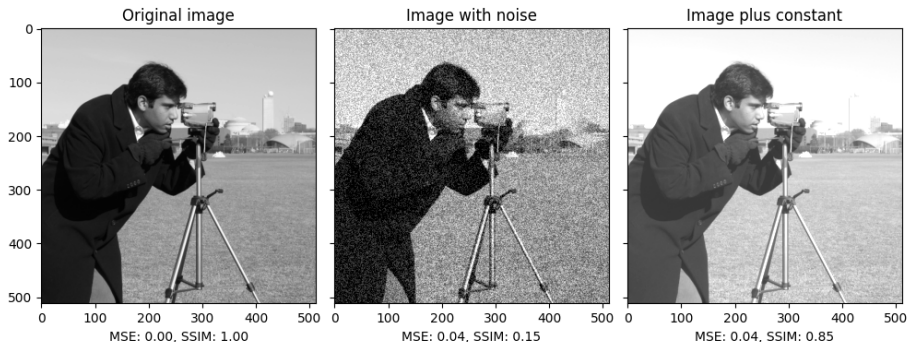
Use SSIM or learned perceptual metrics (LPIPS) for human-aligned quality.

SSIM in words

Structural Similarity Index (SSIM) = type of similarity between two images. Intuitively, prediction of the quality of the second image as a distorted version of the first one, intended as distortion-free



SSIM in words 2



Note: based on the idea that the structural information of an image is more important than the pixel-level information. SSIM compares the structural information of two images by considering the luminance, contrast, and structure

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

with:

- μ_x, μ_y = average pixel intensities of the two images
- σ_x^2, σ_y^2 = variances of the two images
- σ_{xy} = covariance between the two images
- c_1, c_2 = constants used to stabilize the division

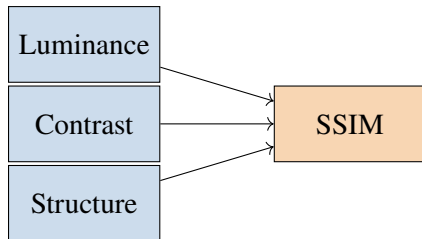
Structural Similarity Index (SSIM)

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Components:

- **Luminance:** Mean intensity comparison
- **Contrast:** Variance comparison
- **Structure:** Covariance (pattern similarity)

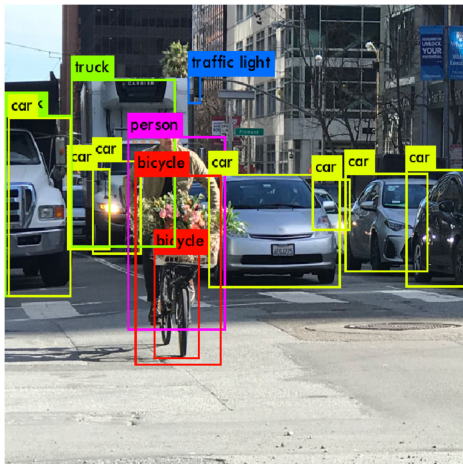
Range: $[-1, 1]$ where 1 = identical



SSIM catches:

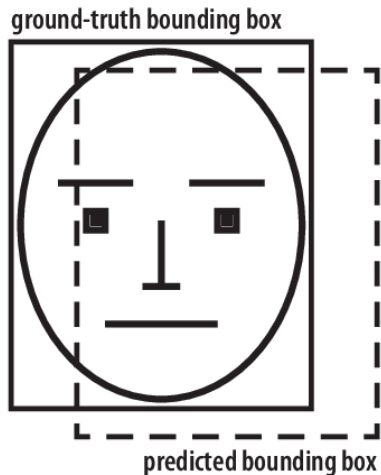
- Blur ✓
- Noise ✓
- Structural changes ✓

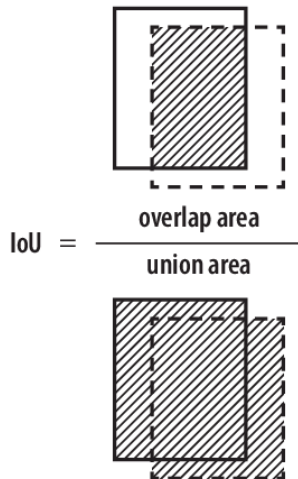
Intersection over Union (IoU) - motivations



wanna detect objects / segmentate an image? Then how accurate are your bounding boxes / segmentation masks compared to the ground truth?

Intersection over Union (IoU) - graphical intuitions



$$\text{IoU} = \frac{\text{overlap area}}{\text{union area}}$$


The diagram illustrates the components of the IoU formula. The top part shows a single rectangle with a shaded 'overlap area' (hatched) and a dashed outline for the 'union area'. The bottom part shows two overlapping rectangles, both shaded with hatching, representing the 'union area'.

Intersection over Union (IoU) is a metric used to evaluate the similarity between two images, especially when one of the images is a ground-truth or a reference image. It is used commonly in object detection and semantic segmentation tasks. It corresponds to a ratio between the number of pixels that are correctly classified as the object of interest and the total number of pixels that are either object of interest or predicted as object

$$IoU = \frac{A_{intersection}}{A_{union}} = \frac{TP}{TP + FP + FN}$$

with:

- TP = number of true positive pixels
- FP = number of false positive pixels
- FN = number of false negative pixels

Perceptual & Semantic Losses

Why Pixel-wise Losses Fail

Problem: MSE in image generation produces **blurry** results.

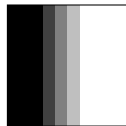
Why?

When uncertain about the correct pixel value, minimizing MSE leads to **averaging**.

Averaging → Blur



Sharp



MSE-optimal

MSE: “I’m uncertain if edge is at position 10 or 12... let’s average and make it gray!”

Perceptual Loss (Feature Loss)

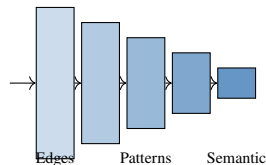
Idea: Compare images in **feature space**, not pixel space.

$$\mathcal{L}_{perceptual} = \sum_i \lambda_i \|\phi_i(\hat{y}) - \phi_i(y)\|^2$$

where ϕ_i = feature map from layer i of pretrained network (e.g., VGG)

Layer choice matters:

Layer	Captures
Early (conv1-2)	Edges, textures
Middle (conv3-4)	Patterns, shapes
Late (conv5, fc)	Semantic content



Perceptual vs Semantic Loss

Perceptual Loss:

“Does it **LOOK** like a real image?”

- Texture quality
- Edge sharpness
- Visual artifacts

Semantic Loss:

“Does it **MEAN** the same thing?”

- Correct objects present
- Right class/category
- Content preservation

Types of Semantic Loss:

Classification-based:

$$\mathcal{L} = CE(Classifier(\hat{y}), target)$$

Embedding-based:

$$\mathcal{L} = \|Enc(\hat{y}) - Enc(y)\|^2$$

Segmentation-based:

$$\mathcal{L} = \sum CE(Seg(\hat{y}), Seg(y))$$

Modern: CLIP loss for text-image alignment

Distribution Distances

$$KL(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

Trap #11: NOT Symmetric!

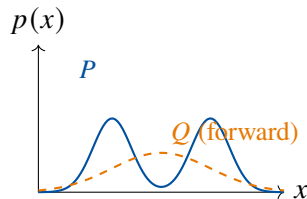
$P = [0.9, 0.1]$, $Q = [0.5, 0.5]$

$KL(P\|Q) = 0.368$

$KL(Q\|P) = 0.511$

Which direction?

- $KL(P\|Q)$: “Forward” — mode-covering
- $KL(Q\|P)$: “Reverse” — mode-seeking (VAEs)



Trap #12

$KL = \infty$ if $Q(x) = 0$ where $P(x) > 0$!

Jensen-Shannon and Wasserstein Distance

Jensen-Shannon Divergence:

$$JS(P\|Q) = \frac{1}{2}KL(P\|M) + \frac{1}{2}KL(Q\|M)$$

where $M = \frac{P+Q}{2}$

Advantages:

- Symmetric
- Bounded: $[0, 1]$
- Always defined

Wasserstein (Earth Mover's):

$$W(P, Q) = \inf_{\gamma \in \Pi(P, Q)} \mathbb{E}_{(x, y) \sim \gamma} [\|x - y\|]$$

Advantages:

- Meaningful when distributions don't overlap
- Considers geometry of space
- Key to Wasserstein GANs

When KL/JS fail

Two non-overlapping distributions:

- $KL = \infty$
- $JS = 1$ (saturated)
- Wasserstein = finite!

Metric Selection Checklist

The 6-Question Framework

1. What's the decision?

Match metric to task type

2. Cost of FP vs FN?

Precision vs Recall, cost matrix

3. Is data imbalanced?

MCC, balanced acc, PR-AUC

4. Output probabilities?

Log loss, Brier, ECE

5. Comparing images?

IoU, Dice, SSIM, mAP

6. Predicting distributions?

KL, JS, Wasserstein

Quick Reference by Task

Task	Primary Metrics	Avoid
Balanced Classification	Accuracy, F1	—
Imbalanced Classification	PR-AUC, MCC, F1	Accuracy, ROC-AUC
Regression with Outliers	MAE, Huber	MSE
Regression without Outliers	MSE, RMSE, R^2	—
Probabilistic Classification	Log Loss, Brier, ECE	Hard accuracy
Segmentation	IoU, Dice, Hausdorff	Pixel accuracy
Object Detection	mAP@various IoU	Single threshold
Image Quality	SSIM, LPIPS	PSNR alone
Image Generation	FID, LPIPS, perceptual	MSE, PSNR
Ranking/Retrieval	NDCG, MAP, MRR	Raw accuracy
Distribution Matching	Wasserstein, JS, KL	MSE on parameters

Goodhart's Law: A Final Warning

*“When a measure becomes a target,
it ceases to be a good measure.”*

Real-world examples:

- Optimizing click-through rate → clickbait
- Optimizing accuracy → ignoring minority classes
- Optimizing BLEU score → grammatically correct but meaningless text
- Optimizing response time → incomplete answers

Takeaway

- **Use multiple metrics**
- **Understand what you're truly optimizing**
- **Always visualize your results**

Summary: Key Traps to Remember

Regression:

- Outliers dominate MSE
- R^2 can be negative
- Same metric, different trust

Classification:

- Accuracy paradox (imbalance)
- Precision-Recall trade-off
- F1 hides information
- ROC-AUC misleads with rare events

Computer Vision:

- PSNR $\neq$ perceptual quality
- IoU misses boundary shifts
- mAP@0.5 vs mAP@0.75 discrepancy

Advanced:

- KL is not symmetric
- KL = ∞ with non-overlapping support
- Easy negatives = no learning

Thank you!

Questions?

TTK 4260: Multivariate Data Analysis and ML
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